LedgerProof Foundation

IN FORMATION · DELAWARE 501(C)(3) · STICHTING EU SUBSIDIARY AMSTERDAM

A structural answer to the GDPR-meets-AI-Act question your published work raises.

An open cryptographic provenance protocol for EU AI Act Article 50 transparency, designed so that the immutability the Act requires does not foreclose the erasability the GDPR requires.

PREPARED FOR Lokke Moerel · Senior Of Counsel · Morrison & Foerster · Amsterdam

PREPARED BY Veronica S. Dawkins · Founder · LedgerProof Foundation (in formation)

DATE 3 June 2026

VERIFIABLE AT `spec.ledgerproofhq.io` · `datatracker.ietf.org/doc/draft-dawkins-scitt-ai-article50`

WHY I AM WRITING TO YOU SPECIFICALLY

Three of your recent publications speak directly to the regulatory tension this protocol is built to resolve.

I am reaching out because the architectural answer described in the next pages is, in substantive terms, the operational response to the question your work raises. I should have led with this in the first message, not with an unframed introduction request.

- **"Do LLMs 'store' personal data?"** · October 2024 — addresses the core question of whether the cryptographic substrate of AI evidence can be considered to contain personal data within the meaning of GDPR Article 4(1).
- **EU Digital Omnibus Amendments to GDPR to Facilitate AI Training** — addresses the regulatory architecture for reconciling AI-development practice with the GDPR's data-minimisation, lawful-basis, and erasability requirements.
- **Your published commentary on data-subject rights under emerging AI regimes** — addresses the operational implications of GDPR Article 15-22 rights as applied to AI-system output evidence.

> **OCPP-AI v1.0** separates the cryptographic-evidence layer from the personal-data layer so that the question your October 2024 paper raises has a structurally negative answer at the substrate: *no*, the immutable anchor does not store personal data, and the erasable receipt store does not store evidence the AI Office cannot independently verify.

THE REGULATORY TENSION

Two regimes, one architectural layer: immutability and erasability appear to be in direct conflict.

GDPR Article 17 — Right to erasure

Personal data must be erasable on legitimate data-subject request. Erasure must be effective, not merely promised. Pseudonymisation is not erasure.

Any architecture in which personal data is published to an immutable substrate is, on first read, in tension with this right.

AI Act Article 50 — Transparency obligations

Evidence of AI-system interaction must survive the deployer's continued cooperation, the vendor's continued cooperation, and any single jurisdiction's hostility.

Operationally this requires immutable, independently verifiable evidence — a property vendor-proprietary logging structurally cannot satisfy long-term.

The default-architecture trap. Vendor-proprietary AI logging satisfies neither regime sustainably: it is erasable only at the vendor's discretion (failing the data subject), and it is preserved only at the vendor's continuing commercial existence (failing the supervisory authority). Both regimes need an architecture that resolves the tension at the layer below either.

THE PROTOCOL'S STRUCTURAL ANSWER

Three layers, separated by design. GDPR mechanics live at the upper two; evidence integrity lives at the lowest.

LAYER 1
SCHEMA

Receipt schema layer — GDPR-by-design

Schema validation *rejects* personal data at write time: email regular expressions, telephone patterns, government identifiers, free-text PII patterns. Fields that could otherwise carry PII (deployer_id, reviewer_role, review_rationale) are bounded to specific identifier formats (LEI, VAT, bounded role vocabulary).

LAYER 2
STORE

Receipt store layer — GDPR-erasable

Each receipt resides in a controller-operated store and is fully erasable per GDPR Article 17. Erasing one receipt invalidates only that receipt's Merkle proof; remaining receipts in the same batch retain their proof integrity. Erasure mechanics are documented, auditable, and binding.

LAYER 3
ANCHOR

Anchor layer — immutable, structurally PII-free

Each batch publishes a fixed 36-byte payload: "LPR1" magic prefix ‖ 32-byte SHA-256 Merkle root. The 32-byte root is cryptographically opaque to underlying receipt content. No party can publish personal data through this layer; the structural exclusion is normative, not policy.

Architecturally, this is the answer to "do LLMs 'store' personal data?" applied to the AI evidence problem: **the immutable substrate does not store personal data, by construction.** The substrate stores integrity commitments. Personal data lives only at the controller-operated layer where Article 17 can act on it.

WHAT THE PROTOCOL IS

OCPP-AI v1.0 — open specification, substrate-neutral, multi-implementation.

- **Specification** Open Cryptographic Provenance Protocol for AI System Outputs (OCPP-AI v1.0). Published under **Creative Commons Attribution 4.0 International**. No patent encumbrance asserted by the Foundation. No commercial licensing.
- **Standardisation track** IETF Supply Chain Integrity, Transparency, and Trust Working Group. Internet-Draft `draft-dawkins-scitt-ai-article50-00` on the IETF Datatracker. Companion arXiv preprint (cs.CR primary classification) submitted today.
- **Substrate-agnostic Anchor Interface** Eight normative properties (I-1 through I-8) defining what any conforming anchor substrate must satisfy. Current reference implementation: **Bitcoin OP_RETURN** (continuous operation since 3 January 2009; satisfies all eight properties). Named alternative: **European Blockchain Services Infrastructure (EBSI)** as the primary EU-sovereign substrate, satisfying seven of eight properties with I-4 under continuing evaluation.
- **Architectural compatibility** W3C Verifiable Credentials 2.0 via published JSON-LD context. eIDAS 2.0 architectural complementarity (not duplication): an OCPP-AI receipt wrapped as a VC may be presented through an established Qualified Trust Service Provider for use cases requiring qualified evidence; OCPP-AI signatures are Ed25519 and do not themselves constitute Qualified Electronic Signatures.
- **Reference implementations** Python, TypeScript, Rust under **Apache License 2.0**. Twenty-nine framework adapters covering principal AI orchestration ecosystems (LangChain, LlamaIndex, OpenAI SDK, Anthropic SDK, Mistral, Cohere, AWS Bedrock, Azure OpenAI, Vertex AI, Hugging Face, and others).

ARTICLE 50 SUB-OBLIGATION COVERAGE

Four of the five operative sub-obligations are addressed in v1.0. Sub-obligation 50(3) is deferred pending GDPR-coordination consultation.

50(1)

Direct user disclosure of AI interaction

ChatbotSession profile. Captures: ai_system_id, deployer_id, deployer_country, disclosure_text_hash, session_start_timestamp. Schema validation rejects PII in disclosure_text values.

50(2)

Synthetic media machine-readable marking

GeneratedContent profile. Captures: content_hash, perceptual_hash, generation_type (AI_GENERATED / AI_MANIPULATED / AI_ASSISTED), source_content_hash, transparency_marker.

50(4)

AI-generated text + human-review exemption

HumanReview profile. Captures: content_hash, reviewer_role (bounded vocabulary), reviewer_organization, review_rationale (bounded vocabulary; PII-rejecting), review_timestamp.

50(6)

Clear & distinguishable manner

transparency_marker field. Present across ChatbotSession and GeneratedContent. Default `"LPR-EU-AI-ACT-50"`; localised values admitted for Member State use.

50(3)

Biometric / emotion-recognition notification

Deferred to v1.1. The notification mechanics interact with GDPR Articles 9 (special categories) and 22 (automated decision-making) in ways that merit dedicated stakeholder consultation before specification, including specifically with privacy counsel in your jurisdiction.

50(5)

Not specified in v1.0

Sub-obligation 50(5) addresses interactive AI categorisation/recommendation transparency; addressed through composition of 50(1) and 50(2) profiles in v1.0.

GDPR CONFORMANCE POSTURE (YOUR DOMAIN)

Five binding architectural commitments. Each is normative in the specification, not policy.

- **Data minimisation by structure (Article 5(1)(c))** Schema bounds field values; bounded vocabularies for reviewer_role and review_rationale; no free-text PII channels permitted at write time.
- **Personal data exclusion at write time** Schema validation MUST reject email-address patterns, telephone-number patterns, government-identifier patterns, and free-text PII regular expressions. A conforming implementation that accepts PII at the receipt schema layer is non-conformant.
- **Erasability preserved (Article 17)** The receipt store is the erasure target. Erasure of a single receipt invalidates that receipt's Merkle proof while preserving proof integrity for other receipts in the same batch. The anchor is unaffected. This separation is essential for joint AI Act / GDPR compliance and is normative.
- **Anchor-layer structural exclusion** The 36-byte anchor payload format MUST NOT permit publication of personal data. Compliance is structural: the 32-byte SHA-256 Merkle root is cryptographically opaque to underlying content. Substrate operators MUST NOT relax this requirement.
- **International transfer posture** Receipts can anchor to EBSI as the EU-sovereign substrate for use cases where Member State competent authorities or controllers require Union-controlled infrastructure. Dual-anchor deployment (Bitcoin + EBSI) is the recommended configuration for deployers under EU jurisdictional reach.

> **Position on the Article 4(1) question.** The Foundation's position, consistent with the architectural separation above, is that the Bitcoin OP_RETURN substrate (and EBSI when conforming) does not *"store"* personal data within the meaning of GDPR Article 4(1), because the anchor payload carries cryptographically no recoverable information about underlying receipt content absent access to the full receipt batch — which lives at the controller-operated, erasable layer.

ENTITY STRUCTURE

Two-jurisdiction Foundation + a separate operating company, designed so the protocol outlives any single entity.

LedgerProof Foundation — United States

Form. 501(c)(3) public charity in formation.

Jurisdiction. Delaware.

Counsel. Adler & Colvin (San Francisco) — specialist 501(c)(3) governance counsel.

Function. IETF specification stewardship, US regulatory engagement, global open-source release of reference implementations.

LedgerProof Stichting — European Union

Form. Stichting in formation under Dutch law.

Jurisdiction. Amsterdam.

Counsel. NautaDutilh (Amsterdam) — specialist Dutch foundation counsel.

Function. EU regulatory engagement, EU enforcement-authority correspondence, EU-resident receipt-anchoring contracts, Member State competent-authority liaison.

LedgerProof Inc. — separate operating entity

Delaware C-corporation, separate from the Foundation. Builds and sells managed services (cloud-hosted receipt anchoring, enterprise support, premium SLAs). Funds the Foundation via a documented grant agreement (€50K/month from August 2026) structured to avoid private inurement per IRS guidance. **No overlapping board members between Inc. and Foundation other than the founding Executive Director through 31 August 2026; full divergence per charter thereafter.**

OPERATIONAL STATE (INDEPENDENTLY VERIFIABLE)

Every claim on this page resolves to a URL or a record you can verify without our cooperation.

- **IETF SCITT Working Group Internet-Draft** `draft-dawkins-scitt-ai-article50-00` live on IETF Datatracker · datatracker.ietf.org/doc/draft-dawkins-scitt-ai-article50/
- **EU AI Office consultation submission** Filed via Futurium 2 June 2026 on the draft Article 50 Guidelines. Bitcoin-anchored as Foundation governance event `gov-2026-06-02-eu-ai-office-consultation`; OpenTimestamps receipt available on request.
- **DG-CNECT memorandum** Delivered 3 June 2026 (today). Stable URL: `spec.ledgerproofhq.io/memos/dgcnect-article-50-reference-architecture-2026-06-03.html`. Anchored as governance event `gov-2026-06-03-dgcnect-memo`.
- **Stibbe Brussels engagement** Engaged 3 June 2026 for institutional desk-officer sourcing at the AI Office and EAIB Secretariat, plus priority EAIB national-rep contacts (FR/DE/NL/EE/IE). Deliverable due 6 June 2026 CET.
- **Reference implementations** Apache 2.0 source: `github.com/vsdawkins-creator/ledgerproof-eu` (public). 29 framework adapters covering principal AI orchestration ecosystems.
- **Specifications + companion artifacts** `spec.ledgerproofhq.io` — OCPP-AI v1.0 core specification, Anchor Interface Specification v1.0, W3C VC 2.0 JSON-LD context, conformance test vector index, anchor substrate registry, governance event records.

WHAT I AM HOPING YOU MIGHT CONSIDER

Two warm introductions in support of the Foundation's pre-launch EU counterweight slate, ahead of 6 July public launch.

Ask 1

Sirma Boshnakova

Member of the Board of Management · Allianz SE · Munich

Exploratory 30-minute call → publishable **Letter of Intent to Evaluate**: a twelve-week structured evaluation of the protocol against Allianz's Article 50 use cases.

Indicative commitment on conversion: approximately €400K covering evaluation scope.
Strategic value to LedgerProof: highest-yield insurance-sector counterweight anchor available for the 6 July launch press kit.

Ask 2

Pieter van der Does

Co-founder & Chief Executive Officer · Adyen N.V. · Amsterdam

Exploratory 30-minute call → **Founding Member anchor conversation**: structured commitment with reciprocal API endpoint integration.

Indicative commitment on conversion: approximately €300K plus warrant plus reciprocal API endpoint commitment.
Strategic value to LedgerProof: Adyen as anchor establishes payments-sector Article 50 implementation pattern.

> If a 20-30 minute call with me first would help you decide whether either ask is appropriate to forward, I would welcome it at your convenience. If you would prefer to start with the technical materials, the OCPP-AI v1.0 core specification and the DG-CNECT memorandum at `spec.ledgerproofhq.io` are the primary substantive references.

DISCIPLINE OF THE ASK

What I am explicitly NOT asking, and why that matters.

- ✕ **Not asking for endorsement of LedgerProof specifically.** The protocol is open, vendor-neutral, multi-substrate. Endorsement-of-vendor is the exact pattern an open-protocol Foundation must avoid.
- ✕ **Not asking for any commercial entanglement.** No fee, no engagement, no equity, no advisory arrangement of any kind. The asks are warm introductions evaluated on their substantive merit.
- ✕ **Not asking you to represent or vouch for technical claims you have not independently verified.** Every technical claim in this document resolves to a URL or a record you can examine directly. The introductions, if you choose to make them, can be limited to facilitation.
- ✕ **Not asking for fast-tracking.** An honest no, or a "not now", is more useful to me than a delay. Both asks have documented backup pathways (Brunswick Group EU as the institutional path to Allianz; Mishi Choudhary as the alternate path to Adyen) which I will exercise without hesitation if either fits poorly with your view.
- ✕ **Not asking for confidentiality beyond ordinary professional norms.** The Foundation's regulatory engagement is, by design, publicly verifiable. The protocol is published openly. Confidentiality applies only to the specifics of any counterparty discussion you facilitate, and is at the level you and the counterparty would set.

CONTACT & VERIFICATION

Every claim in this document is independently verifiable at the URLs below.

AUTHOR	Veronica S. Dawkins · Founder · LedgerProof Foundation (in formation)
EMAIL	`veronica@ledgerproofhq.io`
SPECIFICATIONS	`spec.ledgerproofhq.io` — OCPP-AI v1.0, Anchor Interface, JSON-LD context, memoranda, governance events
IETF	`datatracker.ietf.org/doc/draft-dawkins-scitt-ai-article50/`
SOURCE	`github.com/vsdawkins-creator/ledgerproof-eu` — Apache 2.0 reference implementations
EU CONSULTATION	Submitted via Futurium on 2 June 2026 to the European Commission AI Office on the draft Article 50 Guidelines
DG-CNECT MEMO	`spec.ledgerproofhq.io/memos/dgcnect-article-50-reference-architecture-2026-06-03.html`
COUNSEL OF RECORD	Adler & Colvin (US 501(c)(3) formation) · NautaDutilh (Dutch Stichting formation) · Stibbe Brussels (EU regulatory engagement)

> **This document is internal preparation for your evaluation.** It is not a marketing artifact. The specifications at `spec.ledgerproofhq.io` are the canonical, citable Foundation publications. If anything in this document is inconsistent with the published specifications, the published specifications govern.

Thank you for your consideration, Lokke. If neither ask fits, or if I have misread the appropriate way to approach you entirely, please tell me directly. I would rather know than waste your time.